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// 📝 Mock Practical Exam Paper - Variant 2 (90 Minutes)
// 🔥 Focus: Simple Syntax, Hard Thinking, Out-of-the-box Design, Full OOP Coverage

// ✅ Question 1 - Abstract Class Extension (Chef)

public class Chef extends HotelStaff {
    private String speciality;

    public Chef(String name, String surname, String speciality) {
        super(name, surname);
        this.speciality = speciality;
    }

    public void setSpeciality(String speciality) {
        this.speciality = speciality;
    }

    public String getSpeciality() {
        return speciality;
    }

    @Override
    public String toString() {
        return super.toString() + " Chef - Speciality: " + speciality;
    }
}

// ✅ Question 2 - Duplicate Guest Check

public void addGuest() {
    Scanner s = new Scanner(System.in);
    System.out.println("Enter the first name");
    String name = s.nextLine();
    System.out.println("Enter the last name");
    String surname = s.nextLine();

    for (Guest g : hotelGuestList) {
        if (g.getName().equalsIgnoreCase(name) &&
g.getSurname().equalsIgnoreCase(surname)) {
            System.out.println("Duplicate guest detected. Guest not added.");
            return;
        }
    }

    System.out.println("Enter the room number");
    int roomNum = s.nextInt();
    System.out.println("Enter the number of nights stayed");
    int nightsStayed = s.nextInt();
}
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        Guest guest = new Guest(name, surname);
        guest.setRoomNumber(roomNum);
        guest.setNightsStayed(nightsStayed);
        hotelGuestList.add(guest);
    }

// ✅ Question 3 - Search by Room Number

public Guest getGuestByRoom(int roomNum) {
    for (Guest g : hotelGuestList) {
        if (g.getRoomNumber() == roomNum) {
            return g;
        }
    }
    System.out.println("Room vacant.");
    return null;
}

// ✅ Question 4 - GUI Button: Longest Staying Guest

JButton maxStayButton = new JButton("Longest Stay");
maxStayButton.addActionListener(e -> {
    Guest maxGuest = null;
    int maxNights = 0;
    for (Guest g : guestList) {
        if (g.getNightsStayed() > maxNights) {
            maxNights = g.getNightsStayed();
            maxGuest = g;
        }
    }
    if (maxGuest != null) {
        JOptionPane.showMessageDialog(this, "Longest Stay: " +
maxGuest.getName() + " " + maxGuest.getSurname() + " - " + maxNights + "
Nights");
    }
});
add(maxStayButton, BorderLayout.SOUTH);

// ✅ Question 5 - Runnable Summary

public class StaffSummaryRunnable implements Runnable {
    private ArrayList<HotelStaff> staffList;

    public StaffSummaryRunnable(ArrayList<HotelStaff> staffList) {
        this.staffList = staffList;
    }

    public void run() {

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        int managers = 0, housekeepers = 0;
        for (HotelStaff s : staffList) {
            if (s instanceof Manager) managers++;
            else if (s instanceof HouseKeeper) housekeepers++;
        }
        System.out.println("Managers: " + managers);
        System.out.println("Housekeepers: " + housekeepers);
        System.out.println("Total Staff: " + staffList.size());
    }
}

//  Question 6 - File Handling With Filter

public void importGuestsFromFileFiltered(String fileName) {
    try (Scanner sc = new Scanner(new File(fileName))) {
        while (sc.hasNextLine()) {
            String[] parts = sc.nextLine().split(",");
            if (parts.length == 4) {
                int nights = Integer.parseInt(parts[3].trim());
                if (nights > 2) {
                    Guest g = new Guest(parts[0].trim(), parts[1].trim());
                    g.setRoomNumber(Integer.parseInt(parts[2].trim()));
                    g.setNightsStayed(nights);
                    hotelGuestList.add(g);
                }
            }
        }
    } catch (IOException e) {
        System.out.println("Error reading: " + e.getMessage());
    }
}

//  Question 7 - Filter Managers by License Prefix

public List<Manager> getManagersWithXLicense() {
    List<Manager> listX = new ArrayList<>();
    for (HotelStaff s : hotelStaffList) {
        if (s instanceof Manager) {
            if (((Manager) s).getLicenseNumber().startsWith("X-")) {
                listX.add((Manager) s);
            }
        }
    }
    return listX;
}

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//  Question 8 - Custom Exception for Room

public class InvalidRoomNumberException extends Exception {
    public InvalidRoomNumberException(String message) {
        super(message);
    }
}

// Inside addGuest method
if (roomNum < 1 || roomNum > 999) {
    throw new InvalidRoomNumberException("Room number must be between 1 and
999.");
}

//  Question 9 - Sort Staff by Surname via GUI Button

JButton sortBySurname = new JButton("Sort by Surname");
sortBySurname.addActionListener(e -> {
    list.sort(Comparator.comparing(HotelStaff::getSurname));
    tableModel.fireTableDataChanged();
});
add(sortBySurname, BorderLayout.NORTH);

// End of Variant 2 Answers
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